

$$SR = \frac{4W}{nR}$$

4X

FEDERAL UNIVERSITY LAFIA
FACULTY OF SCIENCE, DEPARTMENT OF PHYSICS.
First Semester Examination 2016/2017 Session
Course Title: General Physics Laboratory I (PHY112)
Credit Unit: 1
Time Allowed: 3 hours.

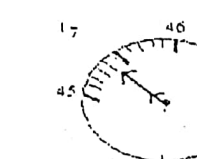
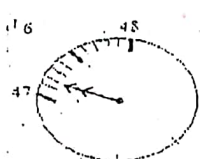
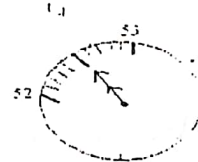
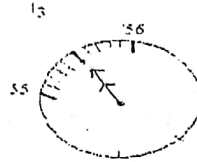
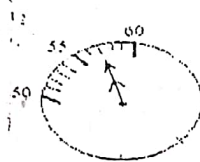
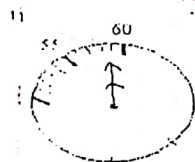
Instruction: Answer all questions in Section A and any two (2) from Section B.

Section A

1. An ammeter with 5 divisions in 1 unit is used to obtain current of $I = 2.5A$. what is the maximum error of the ammeter.
2. The diameter of a wire was repeatedly measured and found to be 0.6mm, 0.14mm, 0.8mm, 0.28mm, 0.19mm and 0.12mm respectively. What is the
 - (i) standard error?
 - (ii) relative error?
 - (iii) percentage error?
3. What are the conditions required of any scientist to enter a laboratory?
4. If $P = 1.52 \pm 0.03$ and $Q = 2.64 \pm 0.02$. Calculate the maximum error if the two errors are maximum errors.
5. Sketch the graph of (i) $y = mx - c$ (ii) $y = -mx + c$

Section B

- Q1. In an experiment to determine the height (H) of a laboratory ceiling using simple pendulum, the following mean time reading (t_1 to t_7) shown below was obtained for twenty (20) complete oscillations at different values of h (h_1 to h_7), as given: 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, and 0.8m respectively. (take $g = 10ms^{-2}$)



$$\frac{1}{2} \frac{1}{N}$$

$$\frac{1}{2N} \approx \frac{1}{2 \times 5} = \frac{1}{10}$$

$$I = 25 \pm 0.1$$

$$\frac{Q}{N} =$$

- (i) Given the relation $T = 2\pi \sqrt{\frac{H-h}{g}}$ solve for $(H-h)$
- (ii) If $\frac{gT^2}{4\pi^2} = y$ and $h = x$. Write the expression for the slope m and the intercept c along the y-axis
- (iii) Plot the graph of $\frac{gT^2}{4\pi^2}$ against h and find the value of H

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- (iv) Determine the slope from the graph
- (v) Determine the standard error on H
- (vi) State two precautions with reason(s) that would be taken during such an experiment
- Q2. In a compound pendulum experiment table I is obtained. Copy and complete the table.

Table I:

h(cm)	Time for 10 oscillations	
	$t_1(s)$	$t_2(s)$
45	16.05	16.34
40	15.89	15.88
35	15.67	15.55
30	15.36	15.40
25	15.23	15.15

- (i) Plot T^2h versus h^2
- (ii) Show from a period of oscillation of a bar pendulum $T = 2\pi\sqrt{\frac{h^2+k^2}{gh}}$ that
- $$T^2h = \frac{4\pi^2h^2}{g} + \frac{4\pi^2k^2}{g}$$
- (iii) From your slope and intercept on T^2h axis, find g and hence k .
- (iv) What does k represent physically?
- (v) Enumerate three (3) types of error and explain any one
- (vi) Show that $P = kQ^n$ can be linear equation if P and Q are the dependent and independent variable respectively.

Q3 In an experiment to determine the specific gravity of a liquid using U-tube.

Using the relation $H + h_l\rho_l = H + h_w\rho_w$

The following results were obtained from the experiment:

$h_l(\text{cm})$	0.65	0.90	1.10	1.20	1.30	1.50
$h_w(\text{cm})$	0.10	0.30	0.40	0.60	0.70	0.80

- (i) In the equation stated above, if $h_l = y$ and $h_w = x$, what is the slope m ?
- (ii) Plot the graph of h_l vs h_w and find the slope S
- (iii) Assume the density of water $\rho_w = 1000\text{kgm}^{-3}$, determine the density ρ_l of the liquid
- (iv) Mention the necessary apparatus and materials required to carried out this experiment
- (v) State two precautions taken to ensure accurate results.

$P = k \dots$

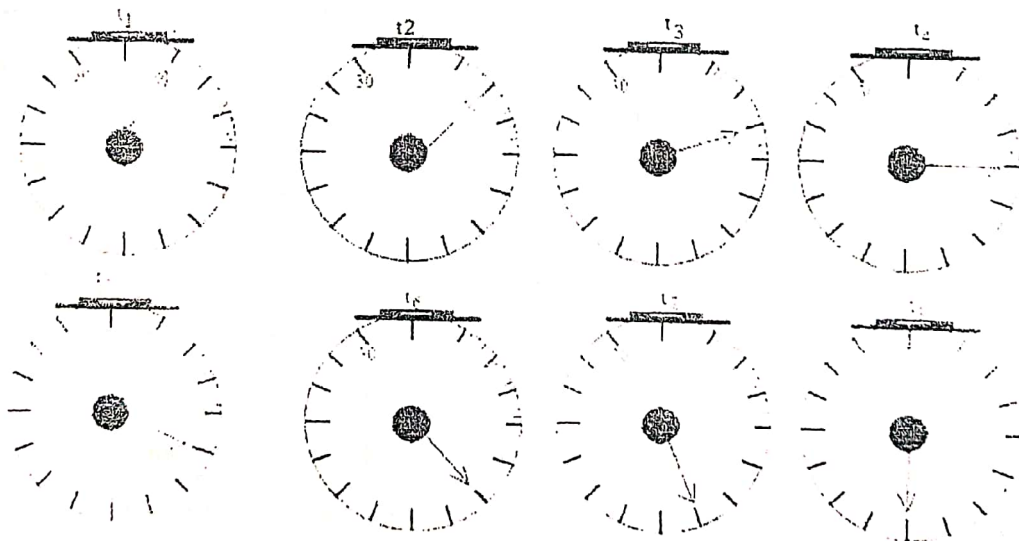
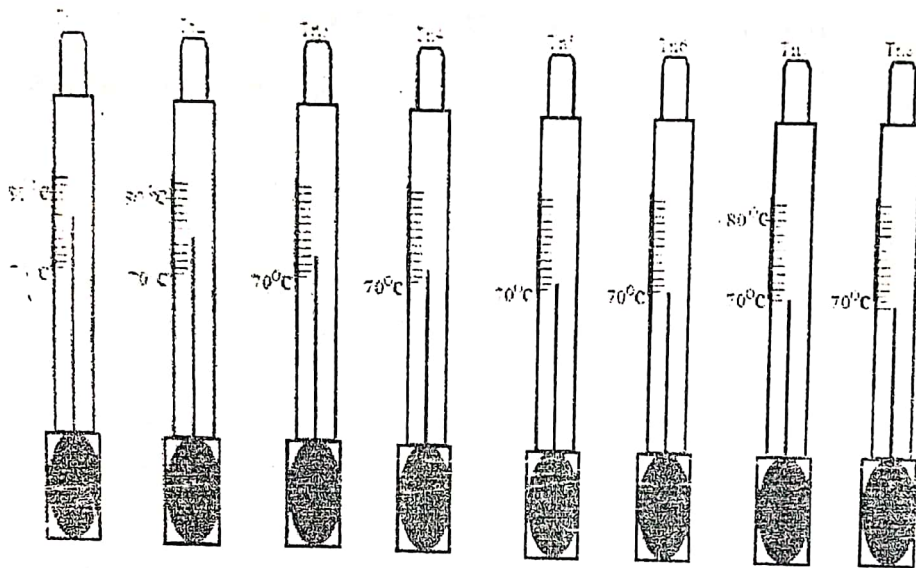
$P = Q^n \dots$



Q2. In an experiment to verify Newton's law of cooling the readings below were taken.

- Use the data to fill the table below and plots the graphs of $T_{th} - T_R$ and $T_{th} - T_{th-1}$ (azad)
- Hence state the Newton's law of cooling.
- State two precautions you would have adopted to ensure accurate readings.

Room Temperature $T_R = 31^\circ\text{C}$



1 unit represents 2 minutes